Name _____ Period _____ Date _____

4-1 Lesson Quiz: Inverse Variation and the Reciprocal Function short form

1. The chart represents an inverse variation. Find the value of p.

x	3	10	15	30
y	5	p	1	0.5

Show your work. The constant of variation is $k =$ _____ because _____

$p =$ _____

2. Write an equation for the inverse variation represented by the table.

x	-3	-1	$\frac{1}{2}$	$\frac{2}{3}$
y	4	12	-24	-18

Ⓐ $y = \dfrac{x}{-12}$ Ⓑ $y = \dfrac{-x}{12}$

Ⓒ $y = \dfrac{12}{x}$ Ⓓ $y = \dfrac{-12}{x}$

Show one product xy that supports your choice: _____

3. Three students can wash a car in 16 minutes. If the time varies inversely with the number of students washing the car, how many minutes will it take two students to complete that same job?

Ⓐ 20 Ⓑ 22 Ⓒ 24 Ⓓ 26

Show your work. Equation: $t =$ _____ $k =$ _____

Scoring: Question 1 and Question 3 are 10 points each. Question 2 is a bonus of up to 5 points. Total out of 20.

4-1 Lesson Quiz, short form: answer key

---

1. $p = 1.5$**.** $k = xy = 3(5) = 15$ (also $15(1) = 15$ and $30(0.5) = 15$), so $10p = 15$ and $p = \frac{15}{10} = 1.5$. Savvas quiz item 1, DOK 1, Realize skill F74. Taught by Example 3 (find k from a known pair, then use it) and rehearsed by the packet's Practice table. **10 points:** 4 for $k = 15$ with a product shown, 4 for $p = 1.5$, 2 for the "because" (the product xy is the same for every pair).

2. D, $y = \dfrac{-12}{x}$**.** Every product is -12: $(-3)(4) = (-1)(12) = \frac{1}{2}(-24) = \frac{2}{3}(-18) = -12$. Savvas quiz item 2, DOK 2, Realize skill F74. Taught by Example 3 Step 1 (k from one pair) and the packet's Practice table (negative k). Distractors A and B are direct variations (y a multiple of x); C has the wrong sign. **Bonus 5:** 5 for D with one product shown; 3 for D alone.

3. C, 24 minutes. With n students, $t = \frac{k}{n}$; $16 = \frac{k}{3}$, so $k = 48$; $t = \frac{48}{2} = 24$. Savvas quiz item 3, DOK 1, Realize skill F74. Taught by Example 3 (the same two steps in a context) and Try It 3. **10 points:** 4 for the equation $t = \frac{48}{n}$ with $k = 48$, 4 for C, 2 for the work shown ($t = \frac{48}{2}$).

Items left off this form (available for a retake or Form B; both are Example 5, which the fast packet cuts): Savvas item 4, the horizontal asymptote of $y = \frac{1}{x+5} - 4$ (answer $y = -4$, DOK 2, F74); Savvas item 5, $y = \frac{1}{x}$ translated 3 units up and 2 units to the left (answer A, $y = \frac{1}{x+2} + 3$, DOK 2, F74). The TE's Lesson Quiz has no graphs, so nothing was redrawn.

IXL replacement: SmartScore 70 or above on PNY and PXE replaces this quiz with 20/20.